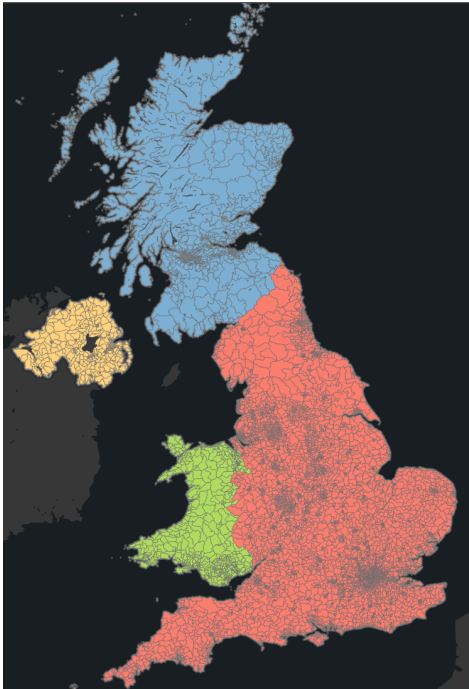


Hello Max,

I would like to report on my progress for the MSc project on the invasion of *C. americana*. I'm refocussing the hypotheses to better delineate and reinforce key aspects, with a revised methodology and its results.

	Before	After
Research question	How are the invasion dynamics of <i>C. americana</i> in the UK mediated both by climate warming trends and urban infrastructure?	
Hypotheses	-H1: <i>C. americana</i>'s spread is shaped by thermal constraints, on two fronts: global climate warming and the urban heat island effect accelerate their proliferation in and around the metropolitan hubs. -H2: Higher human density causes earlier colonisation of areas: jump dispersal is enabled by the inter-city plant trade and the presence of gardens, allowing the beetle to bypass the limitations of diffusion dispersal.	-H1: the initial establishment of <i>C. americana</i> is dictated by a minimum regional climatic temperature threshold. -H2: concentration of beetles in urban hubs increases locally due to human population levels, proxy to both the urban heat island effect, and the presence of their host plants (family Lamiaceae). -H3: the invasion first occurs in the urban hubs, before outskirting to the suburban areas as they become more suitable. -H4: temperature increase predictions will amplify the abundance of <i>C. americana</i> in cities, while enabling previously uncolonised zones to be habitable, creating corridors for its spread.
Datasets & demographic boundaries	-Royal Horticultural Society's Rosemary beetle (<i>Chrysolina americana</i>) monitoring. -Black-headed Cardinal Beetle (<i>Pyrochroa coccinea</i>) occurrences in the UK, on GBIF. -Met Office's Annual Average Temperature Change - Projections (12km).	
	-UKCEH's UK gridded population at 1km resolution for 2021 based on Census 2021/2022.	-2023 Wards for England and Wales. -2022 Intermediate Zones for Scotland. -2021 Super Data Zones for Northern Ireland.
	 a) Grid projection based on 12x12km resolution climate projection from the Met Office.	 b) Census areas (wards in red and green, intermediate zones in blue, and super data zones in orange) of the UK.

Geometry & population delineation	2062 polygons.	9792 polygons: higher sampling size, and finer resolution per polygon on average.
	Squares of equal size (12km x 12km, except on coastline) under the British National Grid coordinate system projection.	Census areas of variable shapes and sizes , which is advantageous in delineating both human and beetle populations: -Urban smaller areas help process high density . -Rural larger areas capture isolated observations .
	Proxy of human population using 1km raster of housing, fit inside each 12km square.	Some census area boundary data don't come with a population value by default, so they have been stitched from other datasets.
Models	Presence model: determining spread, and earliness of colonisation due to temperature (climate and urban heat).	Presence model: determining factors that drive the first detection in an area, like temperature, and adjustment for observer bias.
	Diffusion model: while existent, it isn't the main driver of its distribution in the UK (jump dispersal is). Models predicting a mean continuous expansion rate of x km/year are inaccurate, not representative of the stochasticity of long-distance city-hopping).	Abundance model: as <i>C. americana</i> is already established in the main urban hubs, analysing the evolution of its concentration in already colonised areas to determine a growth rate is a natural next step, and ecologically more pertinent and quantifiable for this invasive species.

Fig. 1: Comparison of previous and current methods and setup.

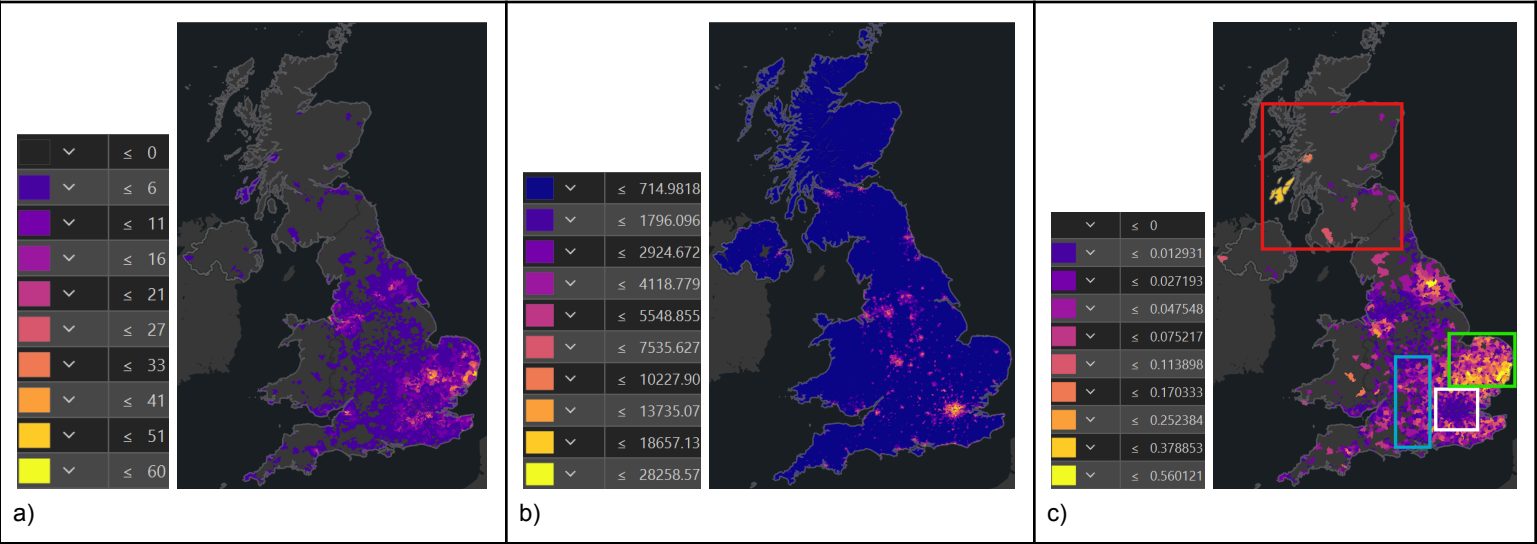


Fig. 2: exemplification of the **observer bias**, proposing that areas with higher human presence also see higher amounts of animal observations, due to higher encounter rates.

- a) *C. americana* observations per census area (mean = 2.86 observations, total = 28039 observations).
b) Human population density/km² per census area (mean = 2627 people/km²).
c) Density coefficient of each census area (mean = 0.007), which follows the formula down below.

$$\text{Density coefficient} = \frac{\text{number of observations}}{\text{population density}}$$

We can also visually represent the effect of observation bias (fig. 2): above are the observations of *C. americana* (a), human density (b), and their ratio (c). The ratio is higher in zones with **more beetle recordings and less humans**, such as semi-urban areas (blue), regions with more focus on wildlife and nature like East Anglia (green), and isolated places where few beetle observations can quickly increase that ratio due to few humans (red). In contrast, the ratio is low in urban centers (white), supporting that more observations in cities are due to **more inhabitants** recording them.

RESULTS

**Table 1: Logistic Hurdle Model
(Presence/Absence Presence Probability)**

Presence Ca			
Predictors	Odds Ratios	CI	p
(Intercept)	0.11	0.11 – 0.11	<0.001
Temp Hist Z	1.49	1.46 – 1.52	<0.001
Presence Pc	2.00	1.72 – 2.33	<0.001
Observations	149100		
R ² Tjur	0.009		

Table 1: Presence probability summary, analysing the correlation of normalised temperature index (Temp Hist Z), and the co-occurrence of the calibration species *P. coccinea* (Presence Pc) using a Generalized Linear Model (GLM).

The model addresses **Hypothesis 1**. It evaluates n = 149100 combinations of ward-year to determine which variables drive a **first detection of *C. americana*** (all with p < 0.001):

- With null predictors (average temperature and absence of *P. coccinea*), the rosemary beetle is **relatively rarely encountered** in any given ward (baseline intercept OR = 0.11).
- There is a positive relationship (Odds Ratio >1.00) between the detection of *C. americana* and the remaining predictors: for every standard deviation of warmth increase in a ward, the odds of at least one observation **increases by 49%** (OR = 1.49).
- If *P. coccinea* was also detected in the same ward-year combination, the odds of observing *C. americana* **doubles** (OR = 2.00). This metric isolates observation bias from the effect of temperature change, using a calibration species.

Table 2: Negative Binomial Count Model (Abundance Index)

Count Ca			
Predictors	Incidence Rate Ratios	CI	p
(Intercept)	0.00	0.00 – 0.00	<0.001
Year	1.28	1.28 – 1.29	<0.001
Pop Density Z	2.20e24	2.86e22 – 1.69e26	<0.001
Temp Hist Z	1.62	1.55 – 1.68	<0.001
Count Pc	0.83	0.77 – 0.90	<0.001
Year × Pop Density Z	0.97	0.97 – 0.97	<0.001
Random Effects			
σ ²	1.94		
τ ₀₀ Zone_Code	0.59		
ICC	0.23		
N Zone_Code	5325		
Observations	149100		
Marginal R ² / Conditional R ²	0.628 / 0.715		

Table 2: Abundance model summary, analysing the multi-variable impact on the proliferation of *C. americana* once it reaches an area. A Negative Binomial Generalized Linear Mixed Model (NBGLMM, type NB2) coupled with random intercepts (locality, or Zone_Code) control for the overdispersal of the count population (variance is much larger than the mean, aka the beetles cluster a lot in select few areas).

The model addresses **Hypothesis 2**. Each predictor is treated in **isolation**, and all hold strong significance (p < 0.001):

- The abundance index **increases by 28% per year** (IRR = 1.28), as more beetles are still colonising the UK territory.
- The Pop Density Z incidence rate ratio of 2.20e24 is a **mathematical artifact** that measures the impact of human density, forced to be calibrated to the year 0. Because of this, we will only keep the direction of the trend, which shows a **positive correlation** between human density and beetle abundance.
- For one standard deviation increase in temperature, beetle abundance **increases by 62%** (IRR = 1.62).
- For every additional record of *P. coccinea*, the observation of *C. americana* **decreases by 17%** (IRR = 0.83), supporting their differences in habitat preference (the former in woodlands, the latter in gardens).
- Year and Pop Density Z interact negatively: population density sees its impact on beetle population **annually decrease by 3%** (IRR = 0.97).

Random effects and predictors variance:

- **Baseline variance** of abundance between census areas (σ² = 1.94).
- Residual, **unexplained variance** within census areas (τ₀₀ = 0.59).
- 23% of the model's **leftover unpredictability** is due to the differences between zones (Intraclass Correlation Coefficient (ICC) = 0.23).
- Analysed zones where *C. americana* occurred (N = 5325).
- Our predictors explain **62.8% of the total variance** in beetle abundance, and **71.5%** if we consider individual census areas.

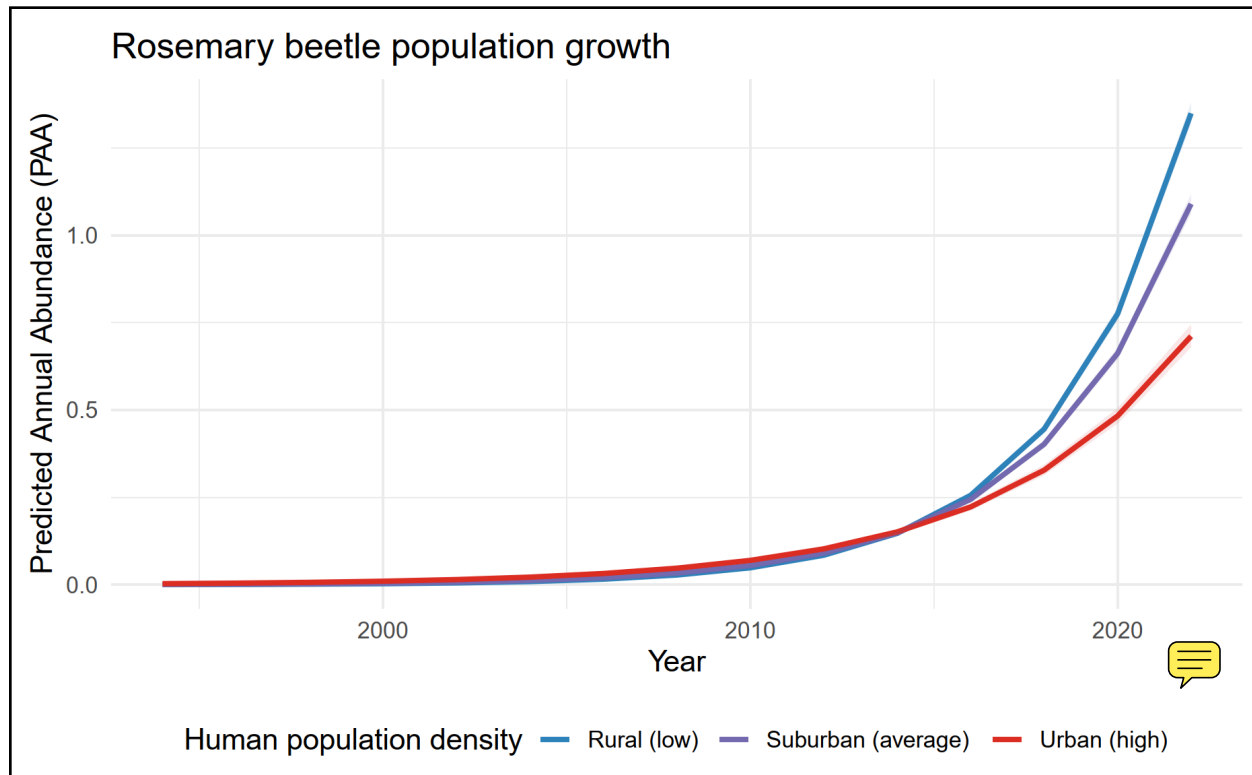


Fig. 3: Predicted annual abundance of *C. americana* over time, in the rural, suburban and urban areas. These three groups are defined with a normalised human population density index (Pop Density Z) of -1 for rural, 0 (average population density in the UK) for suburban, and 2 for urban.

The graph addresses [Hypothesis 3](#). After adjusting for observation bias, the urban, suburban and rural environments were initially seeing an **almost equal increase in beetle abundance**, with urban areas having a slight edge up to 2014. Under the **logistic growth** concept in population ecology, this period corresponds to the **lag phase**, where populations are low and thus reproduction rates do not lift off.

After 2014, the predicted annual abundance (PAA) **increased exponentially faster** in the suburban areas, and even more in the rural ones than in the urban ones. This corresponds to the **exponential growth phase** of a population, where *C. americana* seems to sit nowadays as shown in the graph.

This supports that historically, the introduction and development of *C. americana* in the UK started **primarily in the main urban hubs**. And while it is still increasing there, the beetle found even **greater niche opportunities** in the more garden-rich suburban and rural areas, as it expanded outwards.

However, the IRR value of 0.97 for Year * Pop Density Z (table 2) explains the **slower increase of PAA in urban environments** compared to rural/suburban ones: the positive effect of human population density is **decreased by 3% annually**. Because rural/suburban environments are less tied to human population density, their curves are soaring, being less dampened by this predictor's effect.

At the moment, *C. americana* does not seem to show signs of saturation, but eventually, this will translate into a curve that starts to flatten (also known as the transitional phase), as the populations approach carrying capacity due to resources like host plants getting occupied.

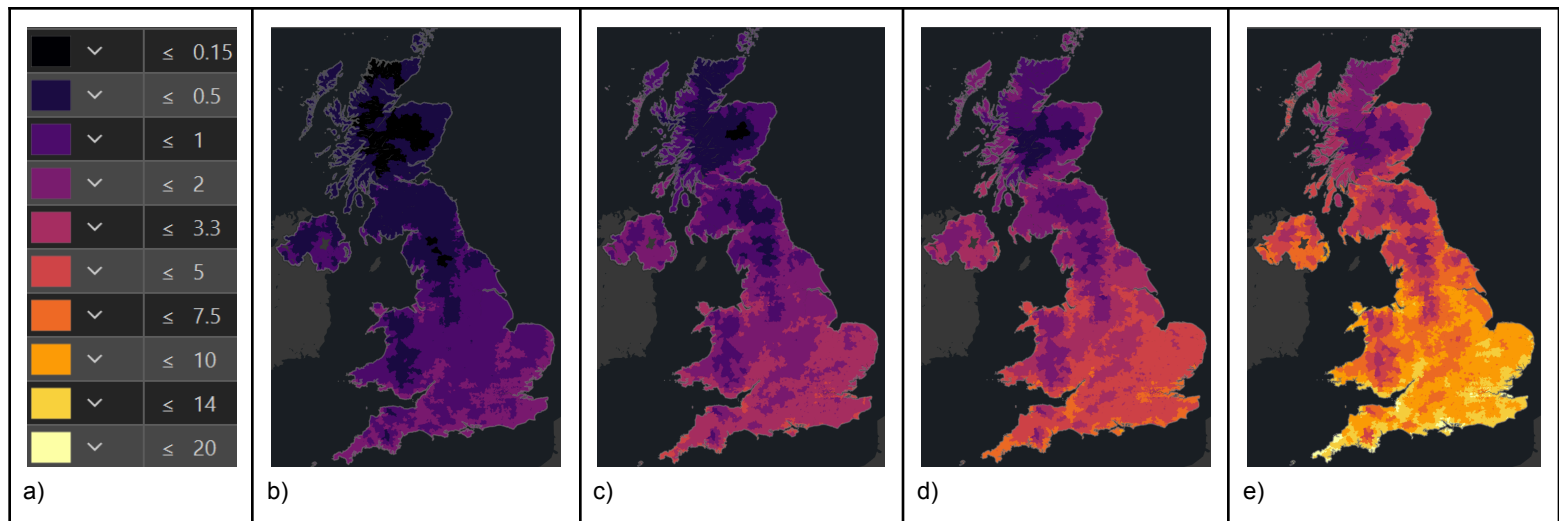


Fig. 4: Predictive spatial modeling of *C. americana* annual abundance across the UK under uniform climate increase, and standardised recording effort throughout the country.

a) Scale legend: predicted relative abundance index, which is the expected average sightings of *C. americana* per year in each ward.

b) Climatic baseline: predicted abundance index using current human population density, and 1981-2000 temperatures.

c-e) Projections under a uniform nationwide temperature increase of (c) 1.5°C, (d) 2.5°C, and (e) 4.0°C above the baseline, and human population held constant.

Interpretation of the maps: for example, in fig. 4c, a census area holding a value of 2.5 (reddish purple covering most of the South) means that for a climate situated +1.5°C above temperatures recorded back in 1981-2000, our model predicts 2.5 beetles to be recorded per year in that particular ward.

In this spatial representation that addresses Hypothesis 4, we focus on the **temperature predictor** (Temp Hist Z, IRR = 1.62) to forecast potential abundance landscapes according to multiple climatic scenarios. In the baseline, the South is the first part of the UK to permit beetle establishment, and increasing temperature scenarios unlock **habitat suitability towards the North**, while **accentuating abundance inside cities**, mostly Southern ones. The brightening of rural space creates potential corridors in which *C. americana* can survive, and even to an extent perform diffusion dispersal.

NEXT STEPS

When the new hypotheses, methods, and results are approved, I will start working on the introduction and discussion parts. As my spatial analysis is complete, I can also insert the maps into ArcGIS Storymaps, which will also serve as the platform for my presentation.

Best regards,

Indi Rick

MSc Taxonomy, Biodiversity & Evolution